

Kalyani Agarwal

Pre-Final Undergraduate @ Amrita Vishwa Vidyapeetham | B.Tech. Artificial Intelligence & Data Science
Major: Artificial Intelligence & Data Science | Minors: Machine Learning & Computer Systems
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EDUCATION

Amrita Vishwa Vidyapeetham, Faridabad
B.Tech. in Artificial Intelligence & Data Science
St. Mary's Senior Secondary School, Jwalapur, Haridwar
Class X: 95.8% | Class XII: 89.4%

2024–2028
CGPA: 9.33/10
CBSE

SCHOLASTIC & RESEARCH ACHIEVEMENTS

- **Patent Inventor – Indian Patent Office, 2026:** Inventor of “Autonomous Robotic Ultrasound System, Methods, and Computer Readable Media Thereof” (Application No. 20261105269A).
- **First Author – STM Journal, 2025:** Published “A Python-Based Investigation of Clinical Data and Ultrasound Images for PCOS Diagnosis” and won **1st Prize** at AI Tech Fusion, Amrita Vishwa Vidyapeetham.
- **First Author – AITA Conference, Bengaluru, 2026:** Presented “Lightweight UAV Avalanche Monitoring Using Monocular Perception and ConvNeXt-Tiny with Attention-Based Classification”; Springer LNCS publication ongoing.
- **International Mathematics Olympiad:** Secured **State Rank 4** in Class XI.
- **Research:** Developing a hybrid SVD+SVM model for Alzheimer's disease classification with spectral and probabilistic enhancements; submitted work on explainable multimodal deep learning for CFRP composites.

RESEARCH EXPERIENCE

IIT Roorkee

Jun'26 – Aug'26

Research Intern – Spatial AI & Multimodal Water Quality Analysis | Guide: Dr. Anshul Yadav

- Processed **4,510+ CWC records** containing **53 water-quality parameters** for the Kanpur–Varanasi stretch of the Ganga River.
- Constructed spatial representations using **500 river reference nodes**, GIS interpolation and spatial feature engineering; fused **5 environmental datasets** into an 18-column multimodal dataset.
- Built a **2-layer GraphSAGE GNN** over a spatial k-NN graph and achieved $R^2 = 0.9783$ with XGBoost.

IIT Tirupati – SEVA Lab

Jun'26 – Jul'26

Research Intern – Computer Vision & Adversarial Robustness | Guide: Dr. Chalavadi Vishnu

- Reproduced and audited the **CVPR 2026 CrowdGen** framework across SASNet and P2PNet using ShanghaiTech Part A/B.
- Implemented a **U-Net perturbation generator with Grad-CAM** under an L_∞ budget of **8/255**.

AICTE × Edunet Foundation × VOIS

Aug'26 – Sep'26

Data Analytics Intern – VOIS for Tech Program

- Selected for a **4-week internship**; applying Python-based analytics across **HR, Agriculture, and Business & Operations**.

SELECTED AI PROJECTS

Fashion Recommendation Platform

2026

Personal Project

- Built a full-stack visual recommendation system using **FastAPI, React, PostgreSQL, Docker, CLIP and FAISS**; processed **44K+ fashion products**.
- Implemented CLIP embedding-based similarity search and an end-to-end image-to-product retrieval pipeline integrating frontend, backend, database and vector search.

MockMate AI – Multimodal AI Interview Coach

2026

College Internship Project

- Developed an AI platform analyzing **audio, video and responses** for communication, body language and answer quality.
- Implemented **6+ multimodal analysis capabilities**: transcription, speaking speed, filler words, eye contact, facial expressions and posture using **Whisper, MediaPipe, React, FastAPI, PostgreSQL and LLMs**.

RecoverAI – AI Revenue Recovery Platform

2026

Razorpay Buildathon – Track 3

- Built an AI-powered payment recovery platform using **FastAPI, PostgreSQL, React and XGBoost**; evaluated **20,000 failed transactions** representing **18.3M+ at risk**.
- Recovered **10.4M+ at 56.80%**, implemented an 8-stage Detect-to-Audit lifecycle, 3-retry caps, deterministic authorization and immutable audit logging.
- Achieved **93/93 backend tests passing** and **100% master validation**.

TECHNICAL SKILLS

Programming: Python, Java, SQL, JavaScript, MATLAB, Data Structures & Algorithms

ML/DL: Scikit-learn, PyTorch, TensorFlow, XGBoost, Random Forest, SVM, CNNs, Transformers, Attention

GenAI: LLMs, RAG, Hugging Face, Embeddings, Prompt Engineering, Multimodal AI, AI Agents

Computer Vision/XAI: OpenCV, YOLOv8, MediaPipe, SHAP, Grad-CAM, Classification, Detection, Segmentation

Data/Systems: NumPy, Pandas, Power BI, Plotly, Feature Engineering, SVD, FastAPI, Flask, REST APIs, React, PostgreSQL, MySQL, MongoDB, Docker, Git, GitHub

LEADERSHIP & ADDITIONAL

- **ORCA Treasures – College AI Club:** Treasurer (2025–Present); managed club expenses, reimbursements and financial records for technical activities and events.
- **Technical Presentation:** Presented MockMate AI to an audience of **500+ students** during the college internship program.